

CS482/495/496 Software Project: Poker Face Project

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1 Client Information

By sharing this client information and the rest of this document, you are stating that this client has provided this project as something they want (not something you created and asked if they wanted), and that they are interested in having you complete this project for your capstone.

- Client name: Dr Hoang Bui
- Client title:
- Client email address:
- Client employer:
- How you know the client:

2 Project Description

2.1 Overview

[Add a few paragraphs describing your project succinctly. What problem are you trying to solve, what is the purpose of your project? Why does your client want this project?]

2.2 Key Features

[At this point you should have a basic understanding of your client's needs. List out the key features of the software system the client wants you to build.]

2.3 Why this Project is Interesting

[Why did you decide this project was interesting enough to you to be a capstone project? What about this project is enticing? Why should anyone care?]

2.4 Areas of CS required

[What subfields of computer science seem most likely to be relevant to your project? A capstone must involve multiple.]

2.5 Potential Concerns and Questions

[Is there any aspect of this project that makes you unsure if it will work, either due to your own interests/background, or that you aren't sure if it fits the requirements? Are there questions you have about this project that you want instructor feedback about?]

2.6 Summary of Efforts to Find a Project

(Not necessary for 482) [Briefly list out when/how you’ve discussed with this client, and if you’ve discussed with other clients who either didn’t work out or didn’t respond. If you considered a different project and it didn’t work out, why didn’t it work out?]

[Most CS495 projects end here. The sections below are for CS482 and CS496 software projects].

2.7 Comparison to Draft

[For CS496 only, focus on highlighting the major differences between the draft proposal in CS495 and this one here. If there are no major differences, you can remove this subsection.]

3 Requirements

3.1 Non-Functional Requirements

[Non-functional requirements are just as important as functional requirements. Dont forget to specify them.]

Table 1 presents the NFRs for this software project.

ID	NFR Title	Category	Description
NFR1	NFR Example 1	Usability	Description of the NFR (it does not follow a user story template)
NFR2	NFR Example 2	Security	Description of the NFR (it does not follow a user story template)

Table 1: Non-Functional requirements

3.2 Functional Requirements (User Stories)

[In CS482, all functional requirements are written as User Stories. In CS496, some projects may use a different template to write the requirements. The table below is an example of writing the Stories. Adapt accordingly to different templates or if you want to display more info.]

Table 2 presents the functional requirements for this software project.

ID	Story Title	Points	Description
S1	Story Example 1	5	As a user, I want to write a user story example, so that people will understand them.
S2	Story Example 2	2	As a user, I want to write a user story example, so that people will understand them.
Total:		80	

Table 2: Functional requirements as User Stories.

3.3 AI Analysis on Functional Requirements

[After you receive your revised User Stories in CS482, or after your first delivery of Stories on CS496, you should use AI as a senior software engineer to help you find any missing or hidden requirements.] [You must provide which LLM you use, which prompt you used, and most importantly **reflect** on the results the LLM gave back. Just because the LLM suggested something, it does not mean it was a valid suggestion and that you should follow blindly. Use your knowledge and reason on which ones are reasonable to add to your requirements and which ones are not.]

4 System Design

4.1 Architecture

We plan to use the Model-View-Controller (MVC) pattern to organize our application. React will handle the user interface, Express and Node.js will handle requests and application logic, and MongoDB with Mongoose will manage our data.

The frontend will communicate with the backend, which will process requests and interact with the database.

Our main modules will be User, Table, Game, Bot, and Chat. The User module will handle accounts, profiles, friends, and admin permissions. The Table module will manage poker tables, while the Game module will handle the rules and gameplay. The Bot module will handle computer-controlled players, and the Chat module will support in-game chat and private messaging.

4.2 Diagrams

[CS482, on sprints/iterations 2-3, you need to create and update a diagram (check Iteration 2-3 assignment for which type of diagram). On CS496, since before sprint/iteration 1, you should have a class diagram and keep it up-to-date. In CS496, if your class diagram changes at each sprint, then create a Class Diagram subsection for the sprints, and show the changes; while keeping the one here the most up-to-date version.]

4.3 Technology

We plan to use the MERN stack (MongoDB, Express.js, React, and Node.js) for our project. JavaScript will be used for both the frontend and backend. React will be used to build the user interface, while Node.js and Express.js will handle the backend. We will use MongoDB with Mongoose for our database and Jest for testing.

4.4 Coding Standards

We will use camelCase for variables and functions, PascalCase for React components and classes, and UPPER_SNAKE_CASE for constants. Our MongoDB collections will also use PascalCase.

We will keep our code formatting consistent and use meaningful names for variables and functions. Comments will be used to explain complex logic when needed.

Each team member will work on separate Git branches, and changes will be reviewed before merging into the main branch. We will use Jest for testing and aim for at least 60% code coverage before merging.

4.5 Data

We will use MongoDB, a NoSQL database, to store our application data. Our main collections will be User, Table, Game, and ChatHistory.

```
User {
  _id: ObjectId
  UserName: string
  Email: string
  PasswordHash: string
  Chips: int
  AdsWatched: int
  Wins: int
  Losses: int
  GamesPlayed: int
}
```

```

    DateJoined: Date
    Avatar: Blob
    IsAdmin: bool
    FriendIDs: ObjectId array
}

Table {
    _id: ObjectId
    JoinCode: string
    Capacity: int
    UserIDs: ObjectId array
    BotCount: int
    IsPublic: bool
    MaxBet: int
    CurrentGameID: ObjectId
}

Game {
    _id: ObjectId
    TableID: ObjectId
    PlayerStates: array
    CurrentTurn: ObjectId
    BettingRound: int
    Pot: int
    CurrentBet: int
    Status: string
}

ChatHistory {
    _id: ObjectId
    TableID: ObjectId (optional)
    UserIDs: ObjectId array
    Messages: array [{
        SenderID: ObjectId
        Words: string
        Timestamp: Date
    }]
}

```

4.6 UI Mocks

4.6.1 Homepage

The homepage introduces the Poker Face platform and provides options to log in, register, or play as a guest.



Figure 1: Poker Face Homepage

4.6.2 Login and Registration

The login and registration pages allow users to create an account, sign in, or continue playing as a guest.

2. Login / Register Page

 **POKER FACE**

[Home](#) [How to Play](#) [About](#) [Contact](#) [Login](#) [Register](#)  Guest

ADVERTISEMENT

Welcome Back!

Log in to continue playing or create a new account.

Login

Username or Email

✉

Password

🔒

☐ Remember me [Forgot password?](#)

Log In

or

 Continue as Guest

Play now without an account.

ADVERTISEMENT

Figure 2: Login and Registration Interface

4.6.3 Lobby

The lobby displays available poker tables and allows players to create, join, or observe games.

6

3. Lobby (Table List)

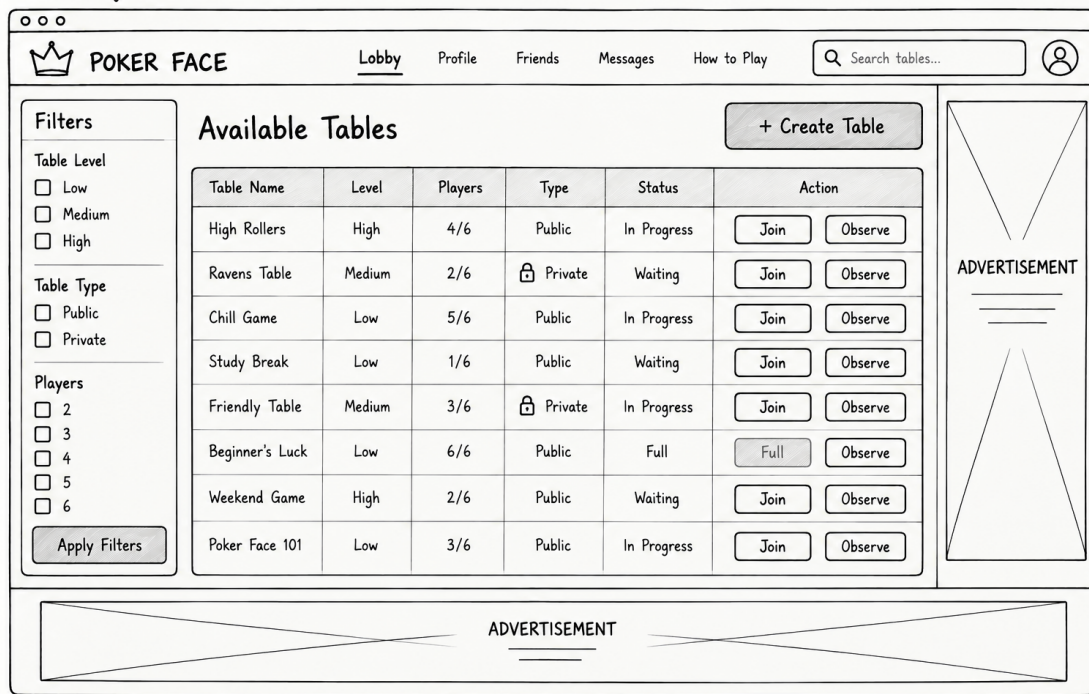


Figure 3: Poker Face Lobby

4.6.4 Table Creation

The table creation interface allows players to configure a table, including its level, capacity, public or private status, and number of bots.

4. Create Table (Modal)

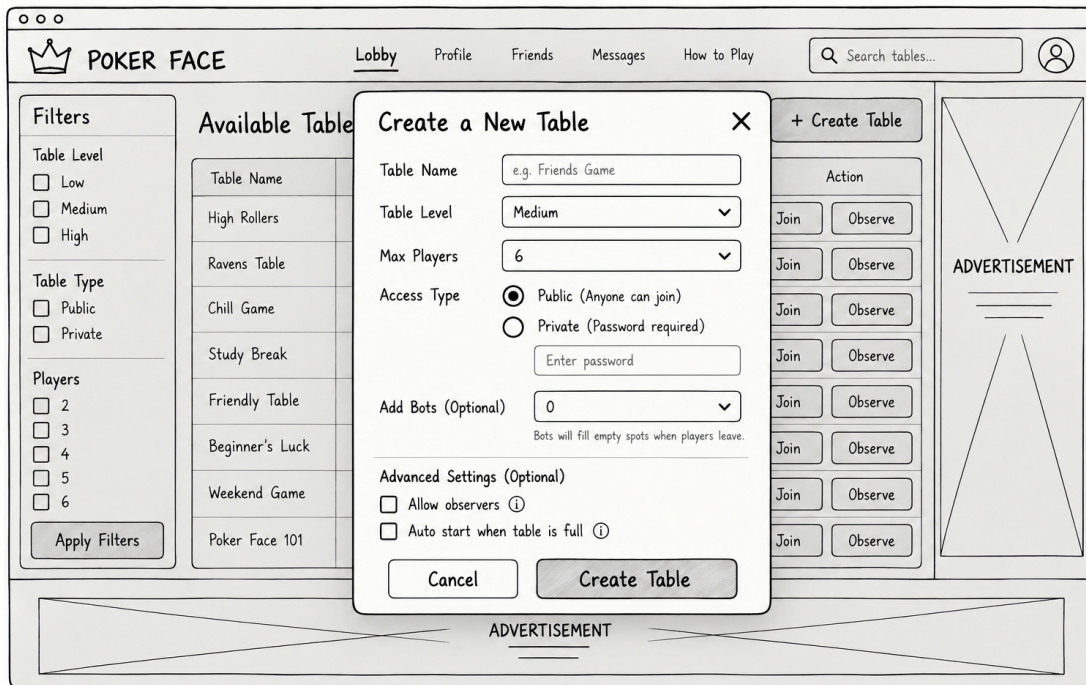


Figure 4: Create Table Interface

4.6.5 Poker Gameplay

The gameplay interface displays the poker table, player cards, chip balances, and betting controls. Players can also communicate through in-game chat.

5. Poker Game (In-Game Play)

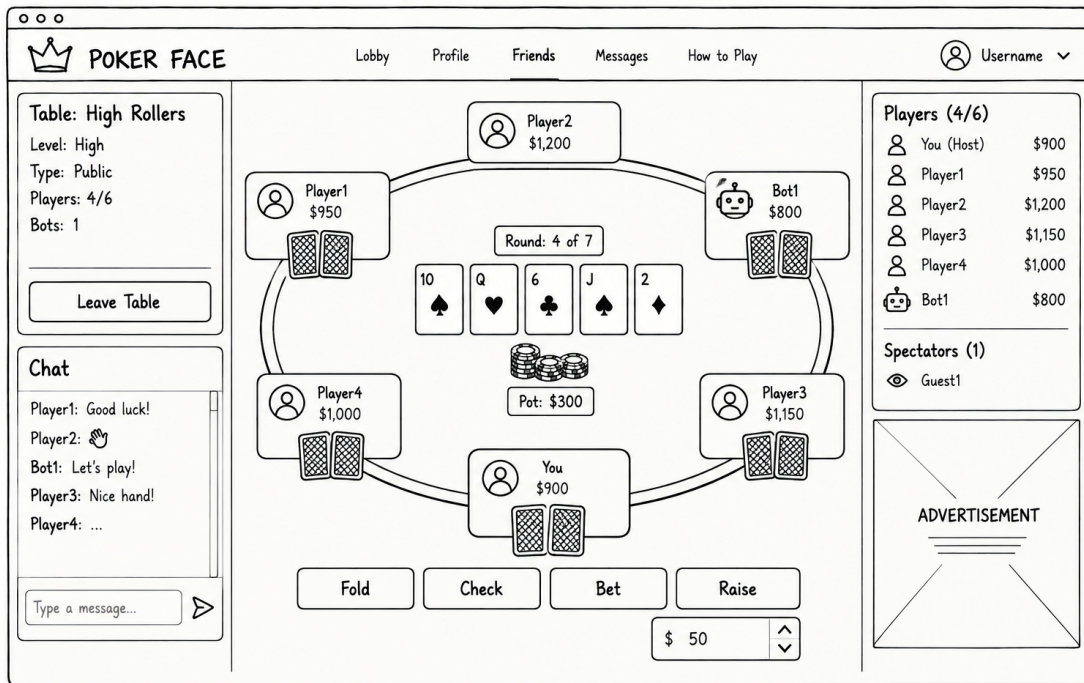


Figure 5: Seven-Card Stud Gameplay Interface

4.6.6 Game Results

The results interface displays the winning player, hand rankings, and chips won or lost after each hand.

6. End of Game / Results

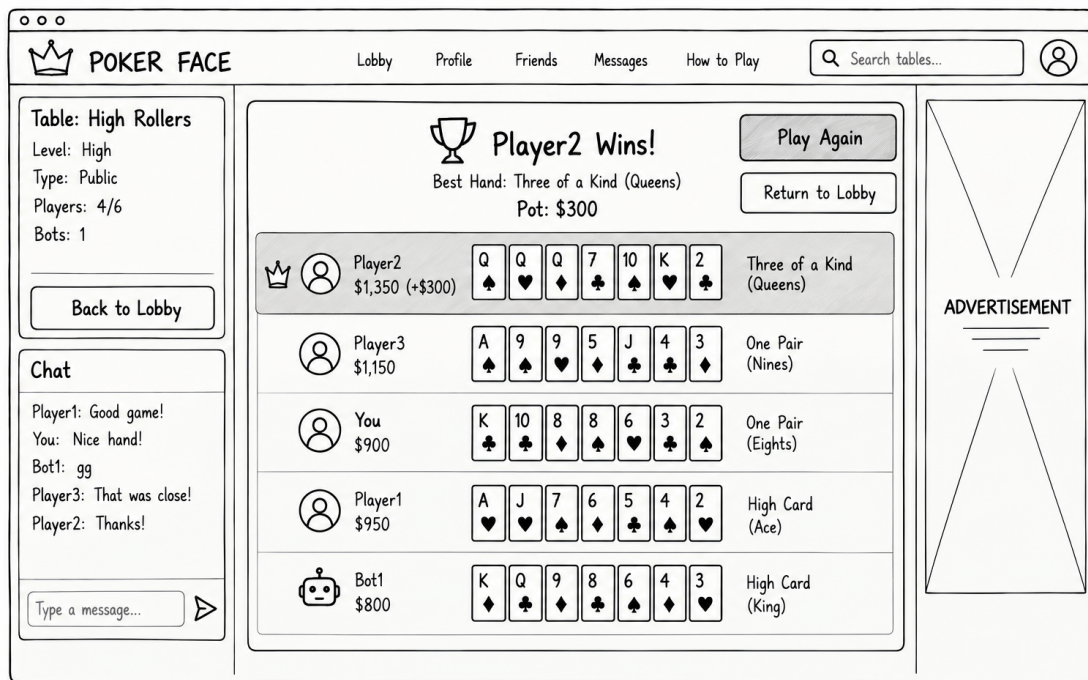


Figure 6: Game Results Interface

4.6.7 Game Tutorial

The tutorial page explains the rules of Seven-Card Stud, including card dealing, betting rounds, and poker hand rankings.

7. How to Play (Tutorial Page)

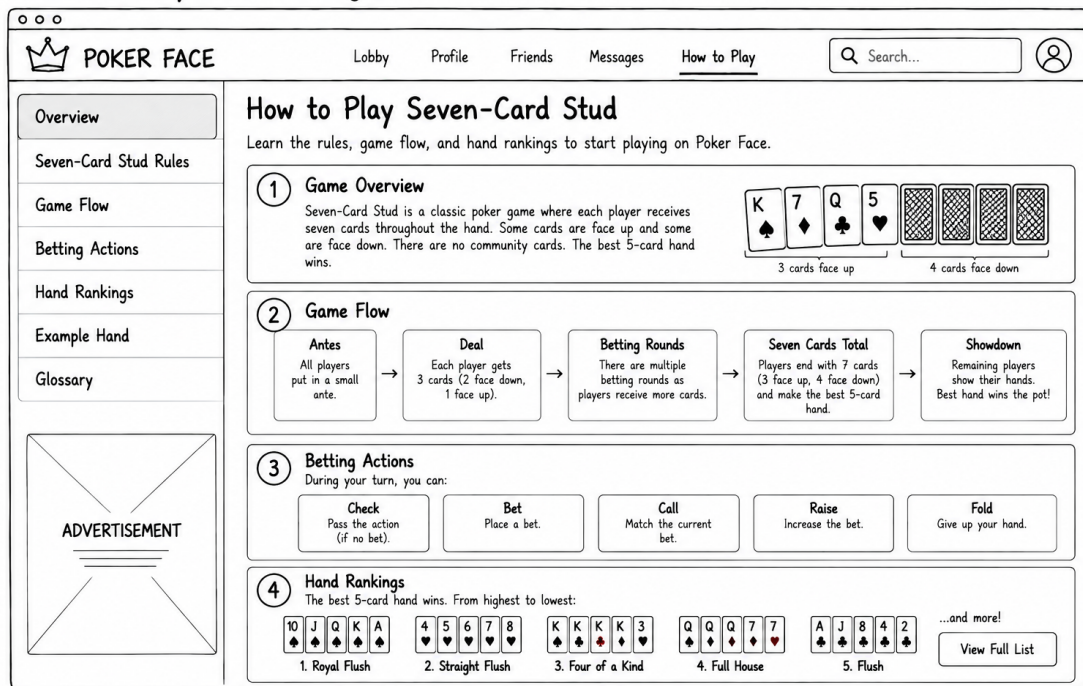


Figure 7: How to Play Tutorial

4.6.8 Player Profile

The player profile displays the user's avatar, chip balance, and gameplay statistics. Registered users can also update their profile information.

8. Player Profile

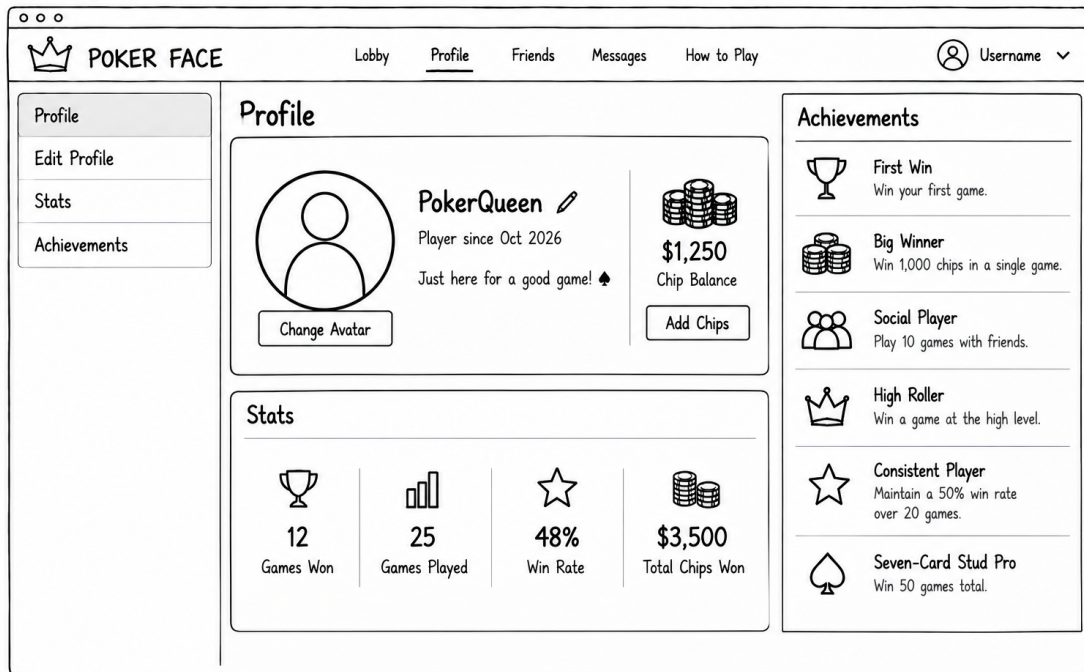


Figure 8: Player Profile Interface

4.6.9 Friends and Messaging

The friends and messaging interface allows registered users to search for players, manage friend requests, and exchange private messages.

9. Friends

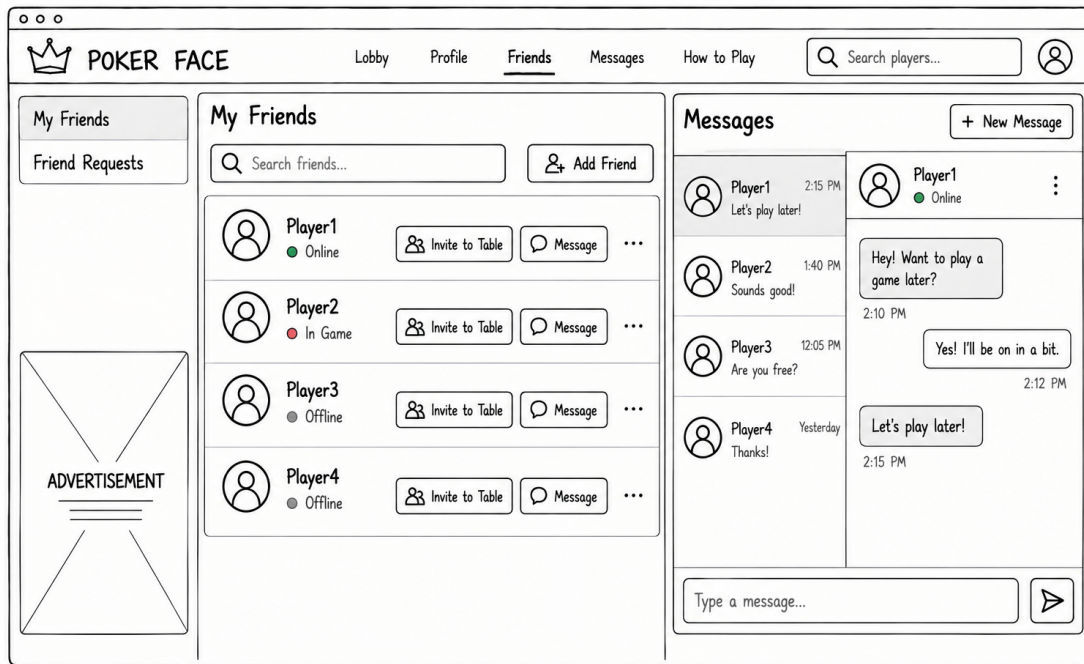


Figure 9: Friends and Messaging Interface

4.6.10 Admin Dashboard

The admin dashboard allows authorized administrators to view registered users, monitor games, and review player statistics and advertising engagement.

10. Admin Dashboard

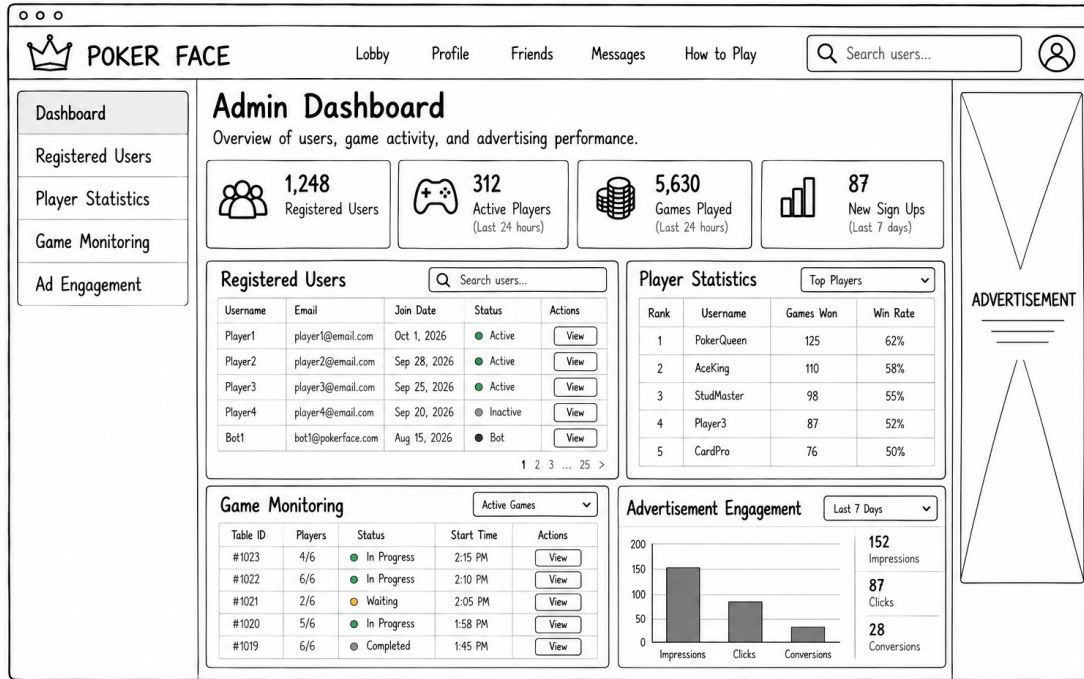


Figure 10: Administrator Dashboard

5 Iterations

5.1 Iteration Planning

[In CS496, you plan all iterations beforehand. In CS482, you update the planning here at each iteration.]

Table 3 shows the iteration planning.

It.	Dates	Stories	Points	
			Planned	Done
1	01/01 - 02/01	S1 Story Example, S2 Story Example 2	07	05
2	02/01 - 03/01	S3 Story Title, S4 Story Title, S5 Story Title, S6 Story Title	17	15
3	03/01 - 04/01	S7 Story Title, S8 Story Title, S9 Story Title, S10 Story Title, S11 Story Title	21	20
4	04/01 - 05/01	S12 Story Title, S13 Story Title, S14 Story Title, S15 Story Title	19	19
5	05/01 - 06/01	S16 Story Title, S17 Story Title	06	06
Total:			70	65

Table 3: Iteration Planning for Incremental Deliveries

5.2 Iteration/Sprint 1

5.2.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.2.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.2.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

Figure 11 shows the test coverage

5.2.4 Retropective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.3 Iteration/Sprint 2

5.3.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.3.2 Work Done

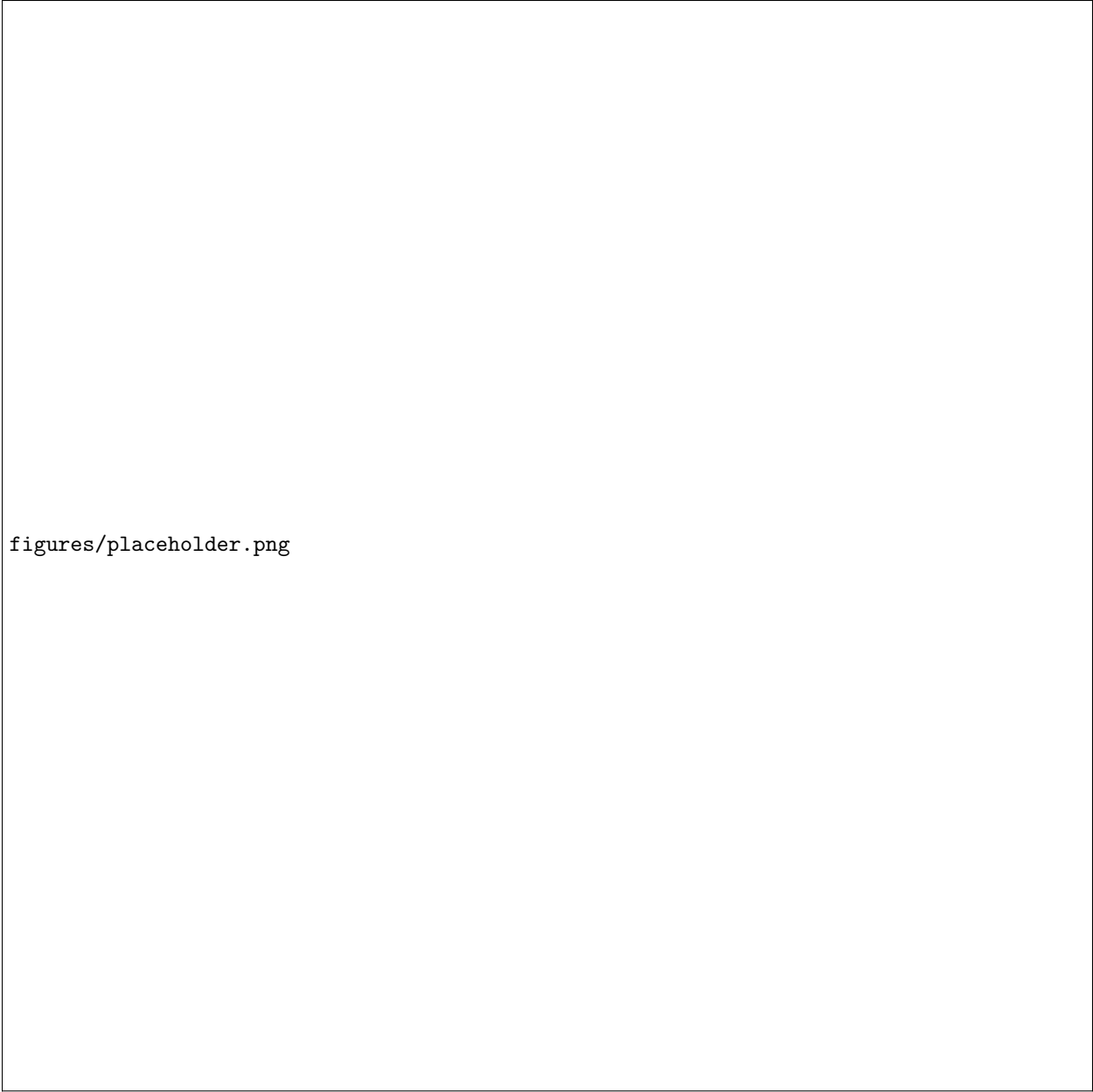
[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.3.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.3.4 Retropective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]



figures/placeholder.png

Figure 11: Iteration 1 test coverage report

5.4 Iteration/Sprint 3

5.4.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.4.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.4.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.4.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.5 Iteration/Sprint 4

[CS496 has 5 sprints. CS482 only has only 3 sprints (remove Iterations 4 and 5 from this doc if you are writing a doc for 482)]

5.5.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.5.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.5.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.5.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.6 Iteration/Sprint 5

5.6.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.6.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.6.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.6.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

6 Final Remarks

6.1 Overall Progress

[Have you completed everything? If so, present evidence on how you brought value to your client, and the overall client satisfaction. Otherwise, estimate how much progress you done and how long it would take to finish this project. Be concrete about your progress, you know how many story points your software is, how many points you completed (this shows your progress). You also how many points your team delivers at each iteration, therefore you can estimate how many more iterations it would take to finish the leftover points (show the math).]

6.2 Project Reflection

[Your personal reflection on the project. What lessons did you learned. What would you have done differently? How can you do better work in future projects? You may write this as a team or per person (or both — if all your iterations were team reflections, then it would be better to write individual reflections here)]

Appendix

[Appendix section if needed]